

Academic Efficiency: Studying Better, Not Just More

FAA - 2 Semester - 25/26 - Grupo x

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Figure 1: Person struggling to study and studying happily (AI Generated), 2026.

Abstract

This project investigates academic efficiency in higher education, with the goal of understanding whether student success depends mainly on the amount of study time or on a broader combination of focus, rest quality, digital habits, and general well-being. Using a multidimensional student dataset, we define two complementary prediction tasks. The regression task aims to estimate students `focus_index`, while the classification task predicts whether a student is likely to achieve a high `exam_score`.

To support these tasks, the data is preprocessed through a structured pipeline that handles missing values, categorical encoding, and feature scaling for mixed-type variables. Baseline interpretable models are first evaluated, followed by more complex approaches and hyperparameter optimization. In addition, unsupervised clustering is used to identify latent student profiles, and model interpretability techniques are applied to explain the most influential factors behind predictions.

The project seeks to provide insight into how study-related and lifestyle-related behaviors contribute to academic efficiency, offering potential value for students, tutors, and academic support services.

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1 Introduction

Academic performance has traditionally been associated with the amount of time students dedicate to studying. However, recent perspectives suggest that academic success is influenced by a broader set of factors, including focus, sleep quality, digital habits, and overall well-being [5]. Understanding how these dimensions interact is therefore essential for promoting more efficient and sustainable learning strategies.

To explore the relationship between these variables and academic outcomes, this project will use of the dataset `student_records_missing.csv`, provided by the course instructors, which contains information about students' study habits, lifestyle, and performance indicators. We defined two complementary prediction tasks: a regression task aimed at estimating the `focus_index`, and a classification task to predict whether a student is likely to achieve a high `exam_score`. A complete machine learning pipeline is implemented, including data preprocessing, baseline model evaluation, advanced model optimization, and clustering analysis.

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Additionally, model interpretability techniques are applied to better understand the factors that most influence predictions, and error analysis is conducted to identify limitations and potential improvements. The goal of this work is not only to build accurate predictive models, but also to derive meaningful insights that can support students, educators, and academic institutions in improving learning outcomes.

2 Problem Definition

2.1 Problem Statement

The main objective of this project is to investigate whether students' academic efficiency depends solely on the amount of time dedicated to studying, or if it results from a combination of factors such as focus, quality of rest, digital habits, and overall well-being. This problem is particularly relevant in modern educational contexts, where students are exposed to multiple sources of distraction and varying lifestyle patterns that may impact their academic performance.

2.2 Target Variables

To address the defined problem, two complementary predictive scenarios are considered.

Regression Scenario: The variable `focus_index` is used as the target for regression, representing a continuous measure of a student's ability to maintain concentration during study activities.

Classification Scenario: The variable `exam_score` is transformed into a binary target representing academic performance. Specifically, a threshold is defined to distinguish between high performance (1) and normal or low performance (0), allowing the problem to be framed as a classification task.

2.3 Stakeholders

Several stakeholders can benefit from the outcomes of this study.

Students: Students can gain insights into whether increasing study time alone leads to better performance, or if other factors such as focus and well-being play a more significant role.

Mentors and Tutors: Educators and tutors can use the findings to recommend more effective and personalized study strategies, focusing not only on study duration but also on behavioral and lifestyle factors.

Academic Support Services: Institutional support services can leverage these insights to design interventions aimed at improving academic success, emphasizing healthy habits, balanced digital usage, and overall student well-being rather than solely increasing study workload.

3 Data Preprocessing Pipeline

3.1 Handling Missing Values

The dataset used in this project contains missing values in both numerical and categorical features, which must be handled before training any machine learning model. Since most algorithms do not natively support missing data, leaving these values untreated could lead to errors or unreliable results, as discussed in the statistical learning literature [4].

Several strategies were considered, such as removing rows with missing values, discarding incomplete features, or relying on algorithms that can tolerate missing data. However, removing rows would reduce the amount of available data and could eliminate relevant patterns, while removing features might discard useful predictive information. For this reason, these alternatives were considered less appropriate for the present scenario.

Instead, a data imputation strategy was adopted in order to preserve the dataset while providing plausible estimates for missing values. The method was defined according to the type of feature:

- **Numerical Features:** Missing values were imputed using the median. This choice is appropriate because the median is more robust to extreme values and skewed distributions, which are common in real-world datasets [4].
- **Categorical Features:** Missing values were imputed using the most frequent category (mode), ensuring that the replacement remains consistent with the observed distribution of each variable.

Although an initial cleaning step was carried out for exploratory and descriptive purposes, the final imputation process used for modelling was implemented inside a preprocessing pipeline with *SimpleImputer*. This ensures that imputation parameters are learned only from the training data during cross-validation and then applied consistently to validation or test data, preventing data leakage. This practice follows standard recommendations for model evaluation procedures [3].

3.2 Categorical Encoding

The dataset contains several categorical variables that must be converted into numerical format so they can be used by machine learning algorithms. Two main encoding approaches were considered: ordinal encoding and one-hot encoding.

Since the categorical features in this dataset do not have a natural order, ordinal encoding would introduce artificial relationships between categories. Therefore, one-hot encoding was selected as the most suitable strategy.

Using *OneHotEncoder*, each category is transformed into a binary indicator variable, avoiding the introduction of any implicit ranking between categories. This approach is consistent with standard practices in statistical learning, where preserving the independence of categorical levels is important for model correctness [4].

In addition, the encoder was configured with `handle_unknown="ignore"`, which improves robustness by allowing unseen categories in validation or test data to be processed without errors.

While one-hot encoding increases the dimensionality of the feature space, this trade-off is acceptable because it preserves the meaning of categorical information and supports the use of a broad range of machine learning models. This effect is related to the challenges of high-dimensional data discussed in the literature [1].

3.3 Feature Scaling

Feature scaling is important to ensure that numerical variables with different ranges do not disproportionately influence the learning process. This is especially relevant for algorithms that are sensitive to feature magnitude, such as support vector machines and

Table 1: Dataset structure before and after preprocessing

Stage	Rows	Columns
Before preprocessing	5000	16
After preprocessing	5000	22

regression-based methods, where distances and coefficient magnitudes directly affect the model [2].

In this project, numerical features were scaled using *RobustScaler*. Unlike standard normalization methods based on the mean and standard deviation, *RobustScaler* centers the data using the median and scales it according to the interquartile range (IQR). This makes it more robust to outliers and better suited to datasets where numerical variables may contain extreme values or asymmetric distributions.

Scaling was applied only to numerical features, while categorical features were left unchanged after one-hot encoding. As with the other preprocessing steps, scaling was integrated into the pipeline so that all transformation parameters are estimated exclusively from the training data. This guarantees methodological correctness, prevents data leakage, and ensures consistency across all cross-validation folds [3].

Overall, the preprocessing pipeline provides a structured and reproducible way of transforming the data, combining imputation, categorical encoding, and robust scaling into a single workflow suitable for fair and reliable model evaluation.

3.4 Comparison of the Dataset Before and After Preprocessing

To evaluate the impact of the preprocessing pipeline, a detailed comparison was conducted between the dataset before and after transformation. This analysis focuses on dataset dimensionality, missing-value treatment, and the structure of the resulting feature space.

Before preprocessing, the dataset contained 5000 instances and 16 features. A total of 2150 missing values were identified across both numerical and categorical variables. After applying the preprocessing pipeline, all missing values were handled through imputation, and the dataset was transformed into a fully numerical representation with 22 features.

The number of instances remained unchanged, indicating that no observations were removed during preprocessing. The increase in dimensionality is explained by the application of one-hot encoding to categorical variables, which expands the feature space in exchange for improved representational capacity [1].

Missing values were present in several variables prior to preprocessing, particularly in *mental_health_score*, *caffeine_intake_mg*, and *screen_time_hours*. After applying median imputation for numerical variables and mode imputation for categorical variables, all missing values were successfully removed.

The categorical variables contain a small number of distinct categories, making them suitable for one-hot encoding.

After preprocessing, categorical variables were transformed using one-hot encoding, resulting in binary indicator features.

Table 2: Missing values before and after preprocessing

Feature	Before	After
gender	150	0
academic_level	100	0
social_media_hours	250	0
screen_time_hours	400	0
caffeine_intake_mg	500	0
mental_health_score	750	0
Total	2150	0

Table 3: Summary of categorical features before preprocessing

Variable	Count	Missing	Unique	Mode
gender	4850	150	3	Male
academic_level	4900	100	3	Postgraduate
internet_quality	5000	0	3	Good

Each encoded feature contains only values in {0, 1}, indicating the absence or presence of a given category. The mode of each variable is 0, reflecting that for each observation, only one category per original feature is active.

After preprocessing, numerical features were scaled using *RobustScaler*.

Overall, the preprocessing pipeline produced a complete, numerically consistent, and machine-learning-ready dataset. The elimination of missing values, combined with appropriate encoding and scaling techniques, ensures that subsequent model evaluation is conducted on reliable and well-structured data.

4 Baseline Models

In this phase, reference models were established to set performance baselines, focusing on interpretable algorithms before transitioning to more complex approaches.

4.1 Regression Task

For estimating *focus_index*, Linear Regression was used as the baseline model. This choice is motivated by the need to understand direct relationships between study habits and focus. As a complement, a Decision Tree (*max_depth*=5) was applied to check for simple non-linear relationships that the linear model might miss. Quantitative comparison validated Linear Regression as the most stable reference model.

4.2 Classification Task

For the classification task (*high_exam_score*), Logistic Regression and a Decision Tree were implemented. Logistic Regression was configured with *class_weight*="balanced" to compensate for class imbalance (approximately 25% positive instances). This model proved to be the chosen baseline, demonstrating superior discriminative capacity compared to the decision tree, which showed signs of overfitting on the training data.

Table 4: Categorical features after preprocessing (one-hot encoded)

Variable	Count	Missing	Unique	Mode
cat_gender_Female	5000	0	2	0
cat_gender_Male	5000	0	2	0
cat_gender_Other	5000	0	2	0
cat_academic_level_High School	5000	0	2	0
cat_academic_level_Postgraduate	5000	0	2	0
cat_academic_level_Undergraduate	5000	0	2	0
cat_internet_quality_Average	5000	0	2	0
cat_internet_quality_Good	5000	0	2	0
cat_internet_quality_Poor	5000	0	2	0

Table 5: Summary of numerical features before preprocessing

Variable	Count	Missing	Mean	Std	Min	Median	Max
age	5000	0	20.520	2.870	16.0	20.000	25.00
study_hours	5000	0	4.540	1.822	0.0	4.530	11.84
self_study_hours	5000	0	2.479	1.178	0.0	2.480	7.41
online_classes_hours	5000	0	2.012	0.984	0.0	2.010	6.00
social_media_hours	4750	250	2.995	1.467	0.0	2.975	8.28
gaming_hours	5000	0	1.565	1.111	0.0	1.490	5.64
sleep_hours	5000	0	7.016	1.164	4.0	7.010	10.00
screen_time_hours	4600	400	6.986	2.476	1.0	6.955	15.30
exercise_minutes	5000	0	74.536	42.932	0.0	75.000	149.00
caffeine_intake_mg	4500	500	251.127	143.810	0.0	251.000	499.00
part_time_job	5000	0	0.498	0.500	0.0	0.000	1.00
upcoming_deadline	5000	0	0.501	0.500	0.0	1.000	1.00
mental_health_score	4250	750	5.505	2.866	1.0	6.000	10.00

Table 6: Numerical features after preprocessing

Variable	Count	Missing	Mean	Std	Min	Median	Max
num_age	5000	0	0.104	0.574	-0.800	0.0	1.000
num_study_hours	5000	0	0.004	0.726	-1.805	0.0	2.912
num_self_study_hours	5000	0	-0.001	0.723	-1.521	0.0	3.025
num_online_classes_hours	5000	0	0.001	0.718	-1.467	0.0	2.912
num_social_media_hours	5000	0	0.010	0.749	-1.558	0.0	2.777
num_gaming_hours	5000	0	0.045	0.665	-0.892	0.0	2.485
num_sleep_hours	5000	0	0.004	0.740	-1.914	0.0	1.901
num_screen_time_hours	5000	0	0.009	0.759	-1.903	0.0	2.666
num_exercise_minutes	5000	0	-0.006	0.572	-1.000	0.0	0.987
num_caffeine_intake_mg	5000	0	0.001	0.623	-1.146	0.0	1.132
num_part_time_job	5000	0	0.498	0.500	0.000	0.0	1.000
num_upcoming_deadline	5000	0	-0.499	0.500	-1.000	0.0	0.000
num_mental_health_score	5000	0	-0.084	0.530	-1.000	0.0	0.800

4.3 Evaluation Metrics

The success metrics were defined to enable rigorous evaluation:

- Regression:** Root Mean Squared Error (RMSE) was the primary metric, due to its stronger penalty for large errors. Mean Absolute Error (MAE) served as a secondary metric for a direct interpretation of average error.
- Classification:** F1-score was selected as the primary metric to balance precision and recall in a class-imbalanced scenario. ROC-AUC was used as a secondary metric to evaluate the model's discriminative ability independent of the decision threshold.

Table 7: Feature composition after preprocessing

Feature group	Number of features
Scaled numerical features	13
One-hot encoded categorical features	9
Total transformed features	22

5 Feature Engineering and Selection

5.1 Overview

Feature engineering and selection aimed to reduce dimensionality while preserving predictive performance. Initial statistical filters (correlation and variance threshold) were applied and complemented with embedded methods based on ensemble models.

5.2 Statistical Pre-screening

Correlation analysis on numerical attributes used an absolute threshold of 0.90 to detect redundancy; no attributes exceeded this limit. Variance Threshold (threshold = 0.01) applied after preprocessing did not remove any features, indicating that the transformed feature space retained sufficient variability.

5.3 Embedded Selection - Classification

For the classification task ('high_exam_score') an AdaBoost-based selector (SelectFromModel with AdaBoostClassifier) was used. The number of features was chosen by nested validation: a GridSearchCV loop with Stratified 15-fold cross-validation optimised the number of selected features. A tolerance of 1% relative to the best mean CV F1-score was used to define a parsimonious choice. The selected subset for classification contains 9 features.

5.4 Embedded Selection - Regression

For regression ('focus_index') selection relied on a Gradient Boosting Regressor inside a SelectFromModel wrapper. The same validation philosophy (KFold with 15 folds and a 1% tolerance on RMSE) led to a parsimonious subset of 6 features for the regression models.

5.5 Remarks

The selected features were dominated by behavioural variables (e.g., study hours, mental health score, sleep hours), while demographic and coarse categorical variables contributed little to predictive power. The reduced feature sets were used in subsequent complex-model training and hyperparameter optimisation.

6 Advanced Models and Hyperparameter Tuning

6.1 Model Selection

We evaluated a set of more complex models to capture non-linear relationships: MLPClassifier, RandomForestClassifier, GradientBoostingRegressor and RandomForestRegressor. Models were trained on the reduced feature sets derived from the feature selection phase.

6.2 Hyperparameter Optimization

Given the computational cost, tuning combined RandomizedSearchCV and GridSearchCV depending on the size of the search space:

- **MLPClassifier:** Randomized search over hidden layer sizes, activation, regularization (alpha) and initial learning rate using Stratified 10-fold CV.
Best configuration found:
– hidden_layer_sizes=(100,)

– activation=tanh

– alpha=0.0018

– learning_rate_init=0.00082
Performance: F1 $\approx$ 0.72.
- **Random Forest (classification):** Randomized search over n_estimators, max_depth, min_samples_split, min_samples_leaf, max_features and class_weight.
Best configuration found:
– n_estimators=500

– max_depth=15

– min_samples_split=5

– min_samples_leaf=4

– max_features=sqrt

– class_weight=balanced
Performance: F1 $\approx$ 0.71.
- **Logistic Regression (tuning):** Grid search over C and class_weight.
Best configuration found:
– C=100

– class_weight=None
Performance: F1 $\approx$ 0.715.
- **Gradient Boosting Regressor:** Randomized search for regression tuning produced the following best parameters:
– n_estimators=300

– learning_rate=0.05

– max_depth=3

– min_samples_split=10

– subsample=1.0
Performance: RMSE $\approx$ 5.71.
- **Random Forest Regressor:** Randomized tuning found the following best parameters:
– n_estimators=500

– max_depth=None

– min_samples_split=5

– min_samples_leaf=1

– max_features=sqrt
Performance: RMSE $\approx$ 5.90.

6.3 Summary

Hyperparameter tuning produced modest but consistent improvements over baseline models. MLP and Random Forest classifiers improved F1 and ROC-AUC slightly, while ensemble regressors provided small RMSE reductions. These results suggest the problem benefits from moderate non-linear modelling capacity without requiring extreme model complexity.

7 Unsupervised Learning (Clustering)

This stage focused on identifying latent subgroups within the student population. The goal was to cluster students with similar

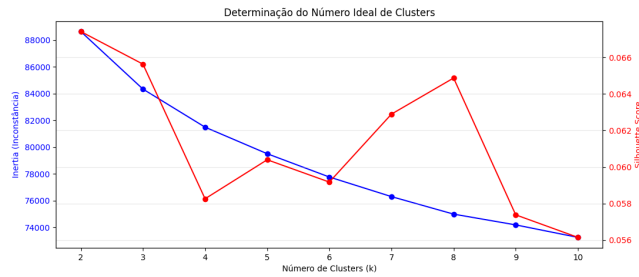


Figure 2: Analysis of cluster centers based on student behaviors. Source: Authors' own work (2026).

behavioral and well-being profiles without using target variables, enabling discovery of patterns based solely on observed attributes.

7.1 Cluster Selection

For K-Means, the data were preprocessed and normalized using StandardScaler to ensure that variable scales do not unduly affect Euclidean distance calculations. The optimal number of clusters (k) was determined by combining two heuristic methods:

- **Elbow Method (Inertia):** Observed the change in inertia as a function of k to identify the elbow point where intra-cluster variance reduction slows down.
- **Silhouette Score:** Computed the silhouette coefficient to measure cluster cohesion and separation.

After visual inspection of Figure 2, $k = 8$ was selected as the optimal number, offering a good compromise between inertia reduction and clear separation among student profiles.

7.2 Cluster Interpretation

Centroid analysis enabled the construction of student "personas" by comparing cluster means to the global population mean. The interpretation revealed distinct behavioral patterns:

- **The Rested Procrastinator (Cluster 0; high: caffeine_intake_mg, sleep_hours; low: study_hours, exercise_minutes):** Sleeps a lot and consumes caffeine but exhibits low study and exercise time.
- **The Zen Athlete (Cluster 1; high: exercise_minutes, mental_health_score; low: social_media_hours, caffeine_intake_mg):** Very physically active, strong mental health, and low social media engagement.
- **The Digitally Exhausted Active (Cluster 2; high: exercise_minutes, screen_time_hours; low: mental_health_score, caffeine_intake_mg):** Physically active but mentally strained by high screen time.
- **The Digital Sedentary (Cluster 3; high: screen_time_hours, social_media_hours; low: exercise_minutes, caffeine_intake_mg):** High engagement with screens and social media, low physical activity.
- **The Connected Veteran (Cluster 4; high: age, social_media_hours; low: exercise_minutes, caffeine_intake_mg):** Older students with high digital activity but low physical activity.

- **The Caffeinated Library Rat (Cluster 5; high: caffeine_intake_mg, self_study_hours; low: upcoming_deadline, screen_time_hours):** Focused on autonomous study and high caffeine intake, with low screen exposure.
- **The Applied Veteran (Cluster 6; high: age, study_hours; low: exercise_minutes, caffeine_intake_mg):** Older students with high study load, sedentary and low caffeine consumption.
- **The Productive Focused (Cluster 7; high: caffeine_intake_mg, mental_health_score; low: sleep_hours, screen_time_hours):** High caffeine intake and limited sleep/screen time but resilient mental health.

These personas demonstrate that the student population is heterogeneous, with distinct behavioral ecosystems that influence academic life independently from direct assessment outcomes.

8 Model Interpretability

8.1 SHAP Analysis

We used SHAP (SHapley Additive exPlanations) to compare local explanations from a tuned Random Forest against a tuned Logistic Regression. TreeExplainer was employed for the Random Forest and LinearExplainer for logistic regression.

8.2 Procedure

Explanations were computed for a selection of test instances representing True Positive, True Negative, False Positive and False Negative cases. For each case we inspected the top contributing features and compared the sign and magnitude of their influence on the predicted probability of high_exam_score.

8.3 Findings

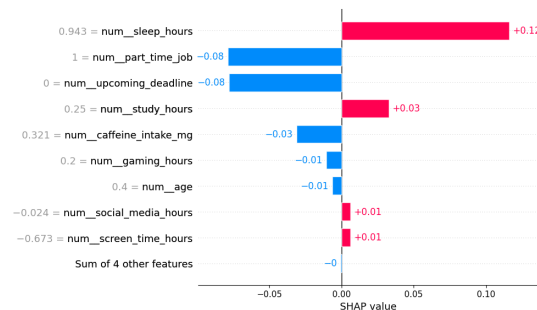


Figure 3: Random Forest False Negative

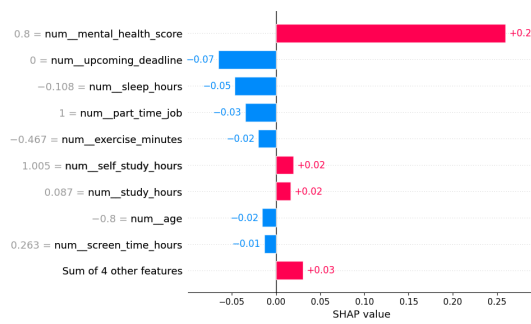


Figure 4: Random Forest False Positive

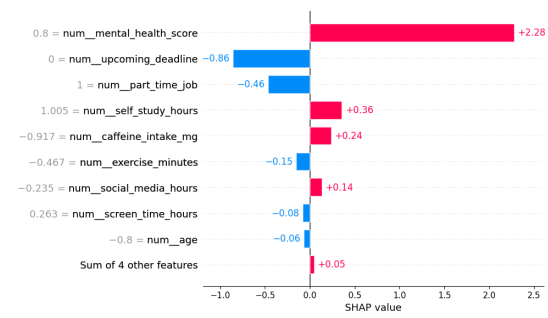


Figure 8: Linear Regression FP

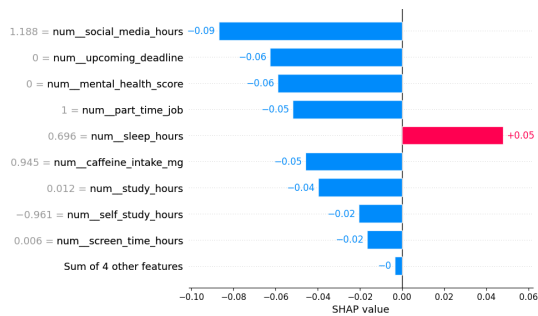


Figure 5: Random Forest False Negative

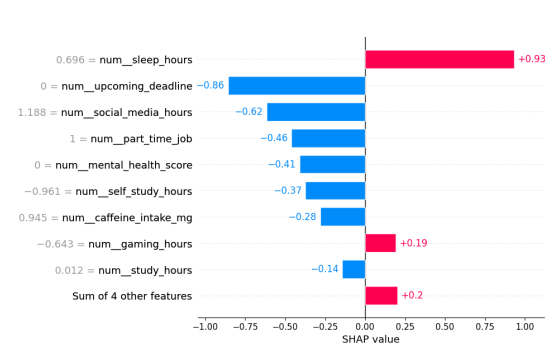


Figure 9: Linear Regression TN

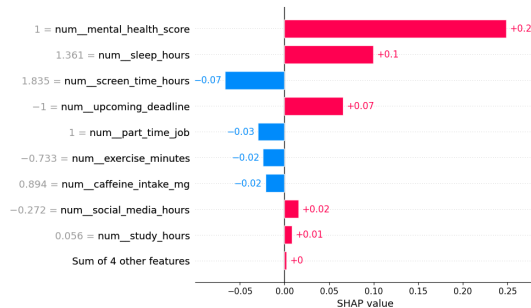


Figure 6: Random Forest TP

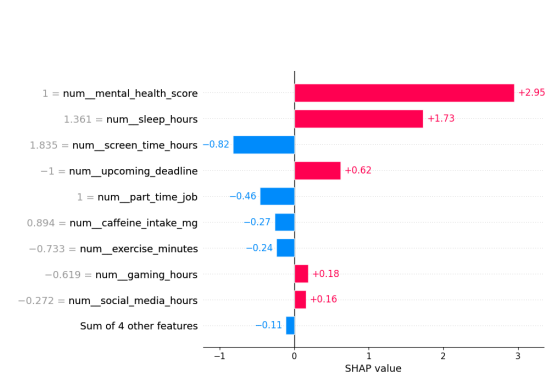


Figure 10: Linear Regression TP

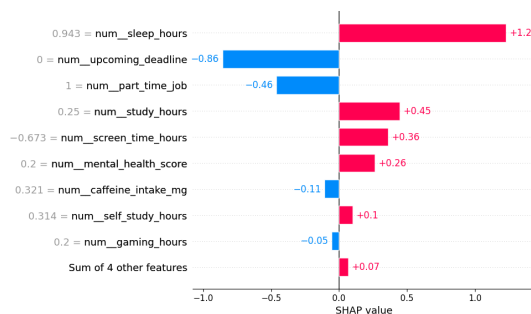


Figure 7: Linear Regression False Negative

- In TP and TN cases there was strong agreement on the top 2-3 influential features (e.g. mental health score, study hours, sleep hours), supporting that these examples are driven by strong linear signals.
- In FP and FN cases the Random Forest often captured interactions that the Logistic Regression missed (e.g. high screen time combined with adequate sleep), explaining some of the discordance between models.
- Scale differences between explanations are expected: Logistic Regression explains variation in log-odds, while tree-based SHAP values relate more directly to class probability shifts.

8.4 Conclusion

SHAP provided actionable insights about when models agree and when complex interactions impact predictions. This comparative analysis supports model selection choices and highlights cases where further feature engineering or alternative modelling strategies could reduce critical classification errors.

9 Error Analysis

9.1 Regression Error Analysis

Regression errors (absolute residuals) were inspected to identify cases with largest deviation. Large errors were often associated with:

- extreme values in predictors (e.g. very high caffeine intake or exercise minutes),
- missing values in relevant features (notably mental health score) where median imputation reduces variance,
- profiles combining high study hours with high distraction metrics (e.g. social media hours), suggesting non-linear interactions not captured by simple linear models.

9.2 Classification Error Analysis

Classification errors showed a predominance of false positives when using class-weighted training (higher recall). Inspection revealed that:

- many false positives are borderline cases close to the threshold of the positive class (quartile-based) and present mixed signals (moderate study but elevated distraction),
- some errors can be attributed to local overfitting in tree-based models where specific combinations of minor features flip predictions,
- missing data in features like `screen_time_hours` contributes to uncertainty in borderline cases.

9.3 Recommendations

To reduce critical errors we recommend: refining imputation strategies (e.g. model-based imputation), exploring interaction features, and calibrating decision thresholds to trade recall/precision according to stakeholder needs.

10 Results and Discussion

10.1 Summary of Findings

Baseline linear models (Linear Regression and Logistic Regression) provided robust and interpretable references. Advanced models (Random Forest, MLP, Gradient Boosting) produced modest improvements after careful tuning.

Key insights:

- Behavioural features (study hours, mental health score, sleep) dominate predictive power for both tasks.
- Demographic features and coarse categorical variables have limited predictive contribution.
- Clustering reveals heterogeneous student personas, suggesting that interventions should be tailored to subgroup characteristics rather than one-size-fits-all recommendations.

10.2 Limitations

The present analysis relies on a single dataset with missing values and potential measurement noise. Although robust preprocessing and cross-validation were applied, external validation on different cohorts would strengthen generalisability.

10.3 Next Steps

Future work includes: exploring richer imputation methods, testing interaction features and higher-order terms, applying calibrated thresholds for classification, and validating findings on external datasets or through targeted educational interventions.

11 Conclusion

At this stage, the first part of the project has been completed, including problem definition, data understanding, and the construction of a preprocessing pipeline suitable for both regression and classification tasks. The results obtained so far show that the dataset was successfully cleaned and transformed into a consistent format for machine learning, ensuring methodological correctness and preventing data leakage. Although the predictive modelling, optimization, clustering, and interpretability stages are still under development, this initial phase already provides a solid foundation for the next steps of the project. The remaining work will focus on evaluating models and extracting meaningful insights about the factors that contribute most to academic efficiency.

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